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RESEARCH ARTICLE

Revolutionizing Mental Health Sentiment Analysis With BERT-Fuse: A Hybrid Deep Learning Model

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ABSTRACT Mental health disorders profoundly affect intellectual ability, emotional resilience, and social stability, leading to lasting effects on both individuals and society. Increasing global concern over mental health calls for innovative solutions, with sentiment analysis emerging as a promising approach for early diagnosis and intervention. Traditional methods are limited by the complexity of mental health-related texts, which contain specialized terminology and domain-specific nuances. To address these challenges, we propose BERT-Fuse, a hybrid deep learning model combining traditional feature extraction techniques (TF-IDF and Bag-of-Words) with advanced neural architectures, including Bidirectional Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Transformer blocks, and Convolutional Neural Networks (CNN). We evaluated three word embedding techniques-BERT, GloVe, and FastText within BERT-Fuse and found that BERT embeddings achieve the highest performance in terms of accuracy, precision, recall, and F1-score. The model was trained on a balanced dataset of text samples representing seven mental health conditions. An ablation study highlights the contributions of key model components to its performance. BERT-Fuse shows promise as a scalable, high-precision system for mental health detection with an impressive average test accuracy of 97.05%. Our model bridges the gap between deep learning and psychological text analysis, paving the way for future mental health assessment tools that are more intelligent and easily accessible.

INDEX TERMS Sentiment analysis, BERT-fuse, hybrid model, hybrid BERT, mental health sentiment analysis.

I. INTRODUCTION

Mental health disorders have become a significant global concern, with recent estimates indicating that over 970 million individuals suffer from conditions such as anxiety and depression (WHO, 2023) [1]. The economic impact is equally alarming, as mental health-related issues contribute to nearly \$1 trillion in global productivity losses annually [2]. Timely detection and intervention are essential to mitigate these adverse effects. The analysis of mental health-related textual

data from social media and online forums presents a promising avenue for understanding emotional and psychological patterns. However, the complexity of linguistic nuances in such texts necessitates advanced sentiment analysis techniques [3].

Sentiment analysis, an essential function in natural language processing (NLP), identifies and classifies emotions inside text [4]. While machine learning models like Support Vector Machines (SVMs) and Naive Bayes are effective at capturing term relevance and frequency, traditional methods like TF-IDF and Bag-of-Words (BoW) struggle with semantic, contextual, and temporal aspects that are important

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for texts pertaining to mental health [5]. Compared to standard approaches, deep learning models, such as Transformer architectures and LSTMs, increase sentiment analysis accuracy by 10-15% [6]. However, there are still issues with domain-specific datasets that incorporate metaphorical language and specialist terminology. These constraints have been addressed with promise by hybrid models that combine deep learning with conventional feature extraction methods like TF-IDF and BoW [7].

To address these problems, we propose **BERT-Fuse**, a hybrid deep learning architecture that integrates BERT embeddings with LSTM, GRU, CNN, and traditional feature extraction methods (TF-IDF and BoW) to improve sentiment analysis precision in the mental health sector. Our method combines stringent evaluation metrics, augmentation, and data pretreatment to guarantee reliable performance. In particular, the model can better comprehend word dependencies and semantic links thanks to BERT embeddings, which capture rich contextual representations of textual data. When LSTM and GRU layers are included, it becomes easier to represent long-range dependencies in sequential text, which makes it especially useful for identifying sentiment patterns over time. By extracting localized features that aid in identifying sentiment-indicative patterns at a finer level, CNN layers further enhance performance. The accuracy of sentiment classification is further strengthened by supplementary feature extraction approaches such as TF-IDF and BoW, which highlight word frequency and importance. Through the integration of these disparate elements, BERT-Fuse successfully closes the gap between conventional and deep learning methodologies, enhancing performance and interpretability.

Key contributions of this paper include:

- We develop **BERT-Fuse**, a hybrid architecture incorporating recurrent layers (GRU and LSTM) with BERT embeddings to capture sequential dependencies in mental health-related texts.
- We integrated CNN layers to extract fine-grained local features for enhanced representation of emotional cues.
- We augment contextual embeddings with TF-IDF and BoW to highlight term-level relevance and frequency patterns.
- We evaluate BERT, GloVe, and FastText embeddings for capturing nuanced sentiment cues, leading to improved classification accuracy.

The remainder of this paper is structured as follows: Section II reviews existing literature on sentiment analysis in the mental health domain. Section III presents the dataset description and data preprocessing. Section IV presents the proposed BERT-Fuse architecture and its implementation. Section V analyses experimental results. Section VI presents a discussion. Section VII explores the study's implications and limitations and potential future research directions, and Section VIII concludes the paper.

II. RELATED RESEARCH

In this section, we summarize the relevant literature, starting with traditional sentiment analysis methods, followed by advancements in deep learning and hybrid models for mental health prediction.

A. MACHINE LEARNING APPROACH

Recent advances in machine learning (ML) have demonstrated the potential to revolutionize mental health diagnostics and interventions through sophisticated data-driven techniques, including sentiment analysis and natural language processing (NLP). Kumar et al. [8] illustrated that ML can effectively reduce mental health stigma by analyzing public attitudes and tailoring personalized educational strategies, while Kaushik et al. [9] highlighted the power of ensemble methods-such as XGBoost-to classify mental health statuses from extensive social media datasets after rigorous preprocessing. Complementing these findings, Suhail et al. [10] combined traditional ML classifiers with advanced pre-trained models like BERT and CatBoost to detect signs of anxiety and depression across diverse online platforms. Additionally, Biró et al. [11] emphasized the role of ML in identifying subtle linguistic markers associated with depression, whereas Shao et al. [12] propose an innovative semantic feature preprocessing approach that mitigates feature sparsity and enhances predictive performance for disorders such as anxiety, borderline personality disorder, and bipolar disorder. Extending the scope further, Alanazi et al. [13] apply deep learning architectures-most notably a single-layer convolutional neural network-to extract sentiment from financial news texts for public mental health monitoring, and Dristy et al. [14] demonstrated that NLP-based classifiers can achieve high accuracy in predicting mental health status from text messages. Moreover, Bouarara [15] and Fatima et al. [16] explored the efficacy of text mining and semi-supervised ML models, respectively, in detecting symptoms of anxiety, depression, and stress, collectively underscoring the transformative role of ML in advancing mental health diagnostics, personalized treatment, and public health surveillance.

B. DEEP LEARNING APPROACHES

Deep learning approaches have emerged as transformative tools in mental health sentiment analysis and monitoring, as evidenced by studies demonstrating innovative architectures and robust performance metrics. For instance, Bao et al. [17] introduced a hybrid model, TCSO-DGNN, integrating a tuned crow search optimized dynamic graph neural network with standard NLP pre-processing techniques to achieve over 93% across precision, accuracy, recall, and F1-scores, thereby enhancing the detection of complex emotional expressions, while Shit et al. [18] compared traditional ML models (logistic regression, XGBoost, SVM, and random forest) with a feed-forward neural network (FFNN), finding that although logistic regression attained 83.07%

accuracy, the FFNN suffered from overfitting, highlighting deployment challenges. Rahman et al. [19] introduced DepressionEmo a dataset of 6037 Reddit posts annotated with depression-related emotions and showed that deep learning models, particularly BERT, outperform conventional methods in multilabel classification. In addition, Zhang et al. [20] employed a hierarchical transformer network with a retrieval-based sampling strategy to achieve 95.46% accuracy in depression risk detection, revealing distinct negative emotional vocabularies among depressed individuals. Further, Kerz et al. [21] enhanced interpretability by combining deep learning models (such as BiLSTM and transformers) with explainable AI techniques like LIME and AGRAD to analyze multifaceted language features, while Oh et al. [22] utilized BERT embeddings alongside LIWC and ANOVA to uncover distinct lexical patterns in Reddit posts for targeted online interventions. Finally, Uddin et al. [23] advanced the application of LSTM-RNNs to detect depressive symptoms, outperforming traditional word frequency approaches on a Norwegian dataset, and Wu et al. [24] demonstrated that integrating word embeddings with a Bi-LSTM-CNN framework effectively identifies emotion labels in psychiatric texts. Collectively, these studies underscore the profound impact of deep learning from transformer-based models to recurrent and convolutional architectures in enhancing the accuracy, interpretability, and applicability of mental health diagnostics and interventions.

C. HYBRID APPROACHES-

Hybrid approaches have emerged as highly effective in mental health sentiment analysis by integrating complementary techniques from both deep learning and traditional machine learning paradigms. Hossain et al. [25] introduced the Opinion-Enhanced Hybrid BERT Model (Opinion-BERT) that employs a multi-task learning framework by integrating opinion embeddings with BERT's contextual representations, achieving a sentiment classification accuracy of 96.77% and a status classification accuracy of 94.22%. Similarly, Karamat et al. [26] proposed a hybrid transformer architecture that combines MentalBERT and MeIBERT with CNN models for mental illness prediction, yielding an overall accuracy and F1-score of 92%. Almutairi et al. [27] presented a hybrid deep learning model that integrates CNN and Two-State LSTM with a feature attention mechanism for predicting depression symptoms from textual data, achieving accuracy, precision, recall, F1-score, and specificity above 97%. Lin and Nuha combined BERT and DistilBERT for text representation with Bi-LSTM and Temporal Convolutional Networks (TCN) for classification, enhancing sentiment analysis in Indonesian texts across diverse topics. Ghosh et al. [28] advanced the field with an attention-based hybrid architecture that leverages a bidirectional LSTM and CNN model to detect depressive sentiments in Bangla social media texts, achieving an accuracy of 94.3% alongside high sensitivity and specificity. Kour et al. [29] employed a hybrid approach

by fusing CNN with bi-directional LSTM to analyze Twitter text data, resulting in a predictive accuracy of 94.28% on a benchmark depression dataset. Finally, Tan et al. [6] proposed a hybrid model that combines RoBERTa and LSTM with data augmentation using GloVe word embeddings to address challenges such as lexical diversity, long-distance dependencies, and class imbalance, achieving F1-scores of 93%, 91%, and 90% across three distinct datasets.

While Hossain et al. [30], Karamat et al. [26], and Tan et al. [6] have advanced hybrid approaches by integrating opinion-enhanced BERT, transformer embeddings with CNNs, and RoBERTa-LSTM respectively, their methods do not fully capture sequential dependencies or augment contextual embeddings with term-level features. In contrast, our BERT-Fuse model leverages GRU and LSTM layers alongside BERT to effectively capture sequential nuances in mental health texts, integrates CNN layers for fine-grained local feature extraction, and further enriches contextual embeddings with TF-IDF and BoW to highlight term-level relevance. Additionally, our comprehensive evaluation using BERT, GloVe, and FastText ensures a robust capture of nuanced sentiment cues, setting our approach apart in enhancing mental health sentiment analysis.

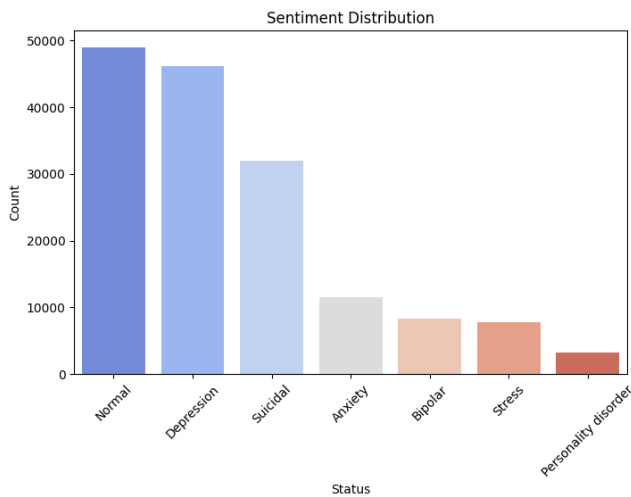
III. DATASET DESCRIPTION AND DATA PRE-PROCESSING

A. DATASET DESCRIPTION

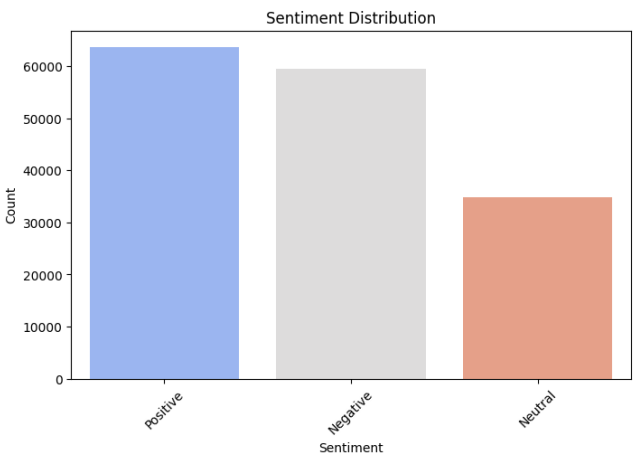
The dataset, sourced from Kaggle, consists of 52,681 text samples capturing mental health-related expressions, where each sample comprises a statement reflecting emotional or psychological states and a corresponding status label indicating the associated mental health condition. The dataset integrated entries from multiple repositories, including conversational data, social media posts, and specialized mental health datasets, ensuring a well-rounded representation across various mental health conditions and contexts. The text samples vary in length and complexity, encompassing colloquial expressions, distress signals, and implicit emotional cues, which introduce challenges such as informal language, misspellings, and nuanced psychological markers. This diverse and large-scale dataset serves as a valuable benchmark for sentiment analysis and mental health classification, facilitating the development of AI-driven early detection systems, psychological assessment tools, and automated mental health monitoring solutions.

We have done Exploratory Data Analysis (EDA) to understand our dataset's main characteristics and identify patterns, trends, and potential issues in the data.

Figure 1 presents the distribution of sentiments in the dataset. Subfigure 1a illustrates the frequency of different sentiment categories. The majority of instances fall under the Normal and Depression categories, followed by Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. This distribution indicates that our dataset contains a higher proportion of general and depressive sentiments compared to less common mental health conditions. Subfigure 1b

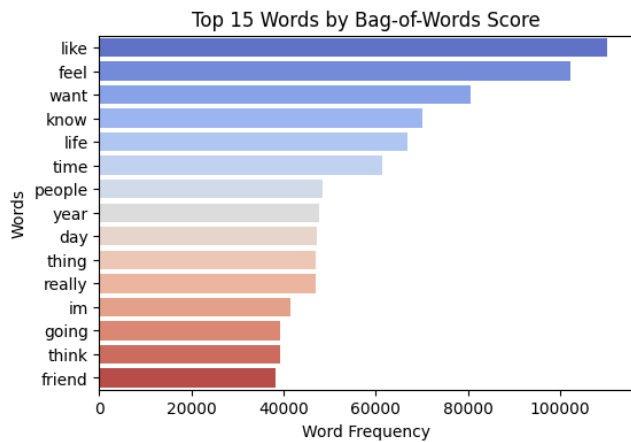


(a) Distribution of mental health status labels, showing the count of instances for various categories such as Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder.

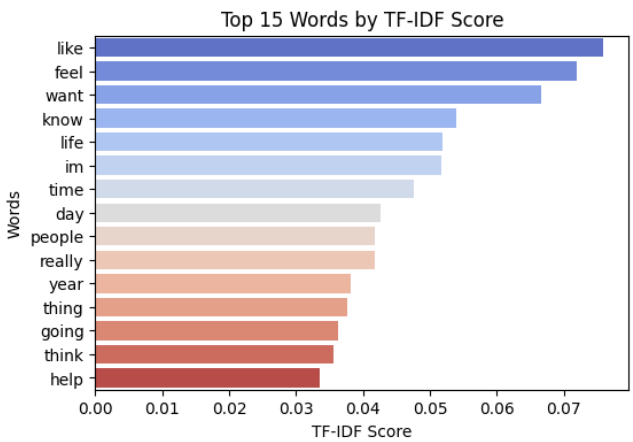


(b) Overall sentiment distribution, categorizing data into Positive, Negative, and Neutral sentiments.

FIGURE 1. Status and sentiment distribution of the mental health dataset.



(a) This bar chart represents the most frequent words in the dataset based on raw word frequency using the Bag-of-Words approach. Common words such as "like," "feel," "want," and "know" appear most frequently, indicating prevalent themes in the dataset.



(b) This bar chart displays the top 15 words ranked by their Term Frequency-Inverse Document Frequency (TF-IDF) scores, highlighting words that are not just frequent but also more informative within specific documents. Compared to Bag-of-Words, this method gives higher importance to words like "help," which may be more relevant to distinguishing different topics.

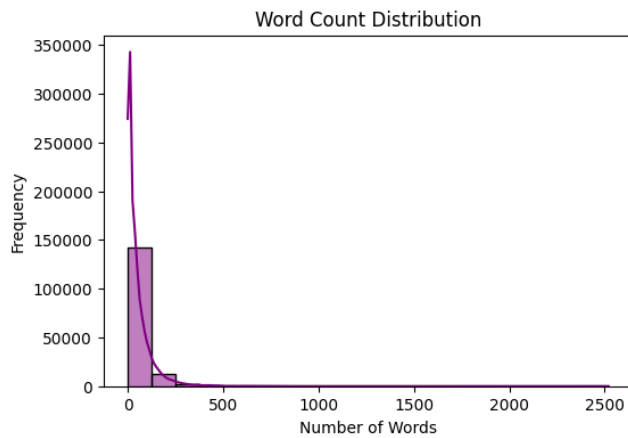
FIGURE 2. Top 15 words in the dataset based on Bag-of-Words and TF-IDF scores.

categorizes these sentiments into broader classes: Positive, Negative, and Neutral. The dataset predominantly consists of positive and negative sentiments, whereas neutral statements are relatively fewer. This suggests that our dataset primarily captures emotionally charged expressions rather than neutral discourse.

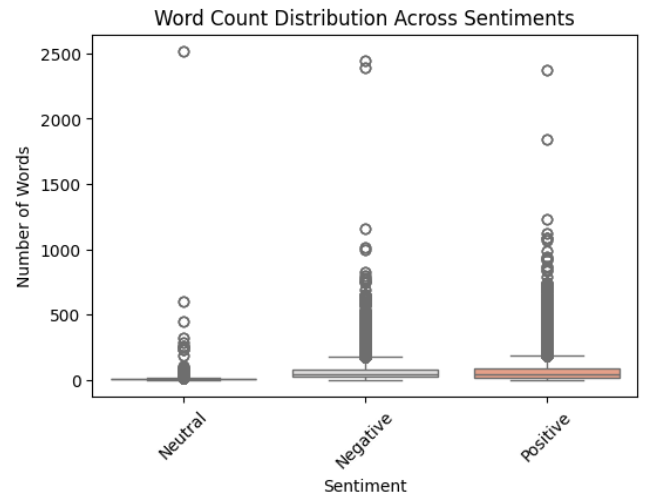
Figure 2 illustrates the most frequently occurring words in the dataset. Subfigure 2a displays the top 15 words based on the Bag-of-Words (BoW) frequency. Common words such as "like," "feel," "want," "know," "life," and "time"

appear frequently, indicating their prominence. Subfigure 2b presents the top 15 words ranked by Term Frequency-Inverse Document Frequency (TF-IDF) score. Unlike BoW, TF-IDF assigns higher significance to words that are particularly relevant in specific contexts but not overly frequent across the dataset. Notably, words such as "help" gain prominence, highlighting their contextual importance.

Figure 3 presents an analysis of word count distribution in the dataset. Subfigure 3a illustrates the overall word count distribution, showing that most text samples contain



(a) This histogram, combined with a density plot, illustrates the distribution of word counts across all instances in the dataset. Most texts contain fewer than 100 words, with a long tail indicating the presence of some much longer texts.



(b) This box plot visualizes the variation in word counts for different sentiment categories (Neutral, Negative, and Positive). The majority of instances have relatively short word counts, while outliers with significantly longer text lengths are present in all sentiment classes.

FIGURE 3. Word count distribution analysis in the dataset.

a relatively small number of words, with a long-tailed distribution indicating a few lengthy instances. Subfigure 3b further examines the distribution of word counts across different sentiment categories. The boxplot reveals that the majority of text samples in all sentiment classes (Neutral, Negative, and Positive) have a limited word count, with a few outliers containing significantly more words.

B. DATA PREPROCESSING

We implemented a complete pre-processing pipeline to ensure that the dataset was ready for analysis and model training.

- **Handling Missing Values:** We resolved missing values by eliminating rows that had incomplete data. This guaranteed that the dataset remained consistent and appropriate for the models, which require all information to perform properly.
- **Handling Class Imbalance:** We rectified the class imbalance in the dataset by integrating weighted measures into the model evaluation procedure. This method guaranteed a fair evaluation of the model's performance by assigning greater weights to underrepresented classes. We mitigated the risk of bias toward the majority class and achieved a more dependable assessment of the model's predictive capability across all classes by examining weighted precision, recall, and F1-score.
- **Selecting Relevant Columns:** To focus on the key aspects of our analysis, we retained only the `statement` and `status` columns. This step removed irrelevant data, reducing noise and simplifying the processing pipeline.

- **Text Cleaning:** We cleaned up the text data by eliminating punctuation and converting all letters to lowercase. This eliminated extraneous symbols and provided homogeneity, allowing for consistent word treatment throughout the dataset.
- **Tokenization:** We split the cleaned text into individual words or tokens. For example, a sentence about emotional anguish was broken down into its constituent terms, such as "restless," "mind," and "month." Tokenization enabled us to conduct a detailed analysis of the text's structure and meaning.
- **Stop Word Removal:** To reduce noise in the data, we eliminated common stop words such as "the," "a," and "is." These words, while common, have minimal semantic value and were removed to increase the emphasis on relevant concepts.
- **Lemmatization:** We used lemmatization to reduce words to their most basic versions. For example, the words "running," "ran," and "runs" were all changed to "run." This phase guaranteed that word variations were treated as comparable, decreasing repetition while maintaining the text's essential meaning.
- **Data Augmentation:** We applied data augmentation to increase the diversity and size of the dataset, using a synonym replacement technique [30]. Let the original dataset be denoted as $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i represents the text data from the "clean_lemmatized_statement" column and y_i represents the corresponding label from the "status" column. For each text entry x_i , we generated $n_{\text{augmentations}}$ augmented versions, denoted as $\{x_i, x'_i, x''_i, \dots, x_i^{(n_{\text{augmentations}})}\}$, using the synonym

replacement function. This function replaces certain words in the text with their synonyms, such as replacing “happy” with “content” or “good” with “great.” As a result, for each original text, we obtained multiple augmented versions, thus expanding the dataset. This augmentation increased the dataset size and provided more varied examples, improving the model’s ability to generalize and reducing the risk of overfitting.

- **Data Split:** We partitioned the dataset into three subsets: 70% for training, 15% for validation, and 15% for testing. This split ensured that the model learned effectively from most of the data while enabling reliable performance evaluation. It allowed us to train the model on a sufficient number of examples, fine-tune hyperparameters using the validation set to prevent overfitting, and objectively assess the final model’s performance with the test set.

C. FEATURE EXTRACTION

- **TF-IDF Vectorization:** The cleaned text was then transformed into numerical vectors using the Term Frequency-Inverse Document Frequency (TF-IDF) technique, which measures the importance of words within a document relative to a collection of documents. A maximum of 1000 features was selected to focus on the most informative terms. Using the training data, we trained the model to learn the vocabulary and IDF weights, which were later used to transform both validation and testing data.
- **Bag-of-Words Vectorization:** We also used the Bag-of-Words (BoW) technique to represent the text in numerical vectors. This method treats a text as a collection of words, preserving word frequency but disregarding grammar and word order. The vocabulary was learned from the training data, and the resulting BoW vectors were used for validation and testing, ensuring the most frequent terms were captured.

D. BERT TOKENIZATION

We used the BERT tokenizer from the Hugging Face Transformers library to achieve a more advanced representation. The tokenizer was trained using the “bert-base-uncased” model [31]. Token IDs and attention masks were used to encode the text into numerical representations. To ensure uniform input dimensions, longer sequences were truncated to a maximum length of 80 tokens, while shorter ones were padded with special tokens at the start and end of each. Attention masks were created to differentiate between meaningful tokens and padding elements, resulting in structured inputs suited for the BERT model’s processing pipeline.

E. SENTIMENT ANALYSIS

We used sentiment analysis on mental health-related utterances to better grasp the emotional context of different

TABLE 1. Sentiment classification breakdown across psychological conditions.

Status	Negative	Neutral	Positive
Anxiety	5264	1302	4957
Bipolar	3150	243	4938
Depression	21563	3504	21145
Normal	8228	26548	14253
Personality disorder	1340	181	1710
Stress	3565	379	3817
Suicidal	16290	2726	12940

psychiatric illnesses. Using the TextBlob library, we classified each sentence as Positive, Neutral, or Negative depending on the emotion polarity. Positive sentiments had a polarity larger than zero, neutrals had zero, and negatives had a polarity less than zero.

We ran sentiment analysis on a collection of statements, each labeled with a psychological status (e.g., Anxiety, Depression, Stress). The model gave a sentiment classification for each statement, which we then classified by status. This resulted in a breakdown of sentiment categories for each condition, demonstrating how frequently each sentiment occurred across various statuses.

Table 1 shows how sentiments differed among conditions by organizing the sentiment counts for each one. For example, Depression had the most negative sentiments (21,563), whereas Normal had more neutral sentiments (26,548). This analysis emphasizes the emotional tone associated with distinct disorders and provides insights into how people with various mental health statuses communicate their feelings, which can help mental health professionals conduct text analysis for therapeutic purposes.

IV. PROPOSED MODEL

Sentiment analysis in mental health demands an architecture that can capture the intricate and nuanced correlations seen in emotionally charged text. To overcome this issue, we created a hybrid model called BERT-Fuse, which combines BERT embeddings with LSTM, GRU, and Transformer blocks, followed by convolution and pooling layers for feature extraction. The pooled features were then processed through dense layers before being concatenated with TF-IDF and BoW representations. Softmax activation was used for the final categorization. Figure 4 illustrates how the model’s architecture optimizes performance by incorporating these components. The following section describes in detail the functioning of each layer in the model.

A. INPUT REPRESENTATION

The pipeline of our model architecture, BERT-Fuse, begins by representing input text sequences using two components:

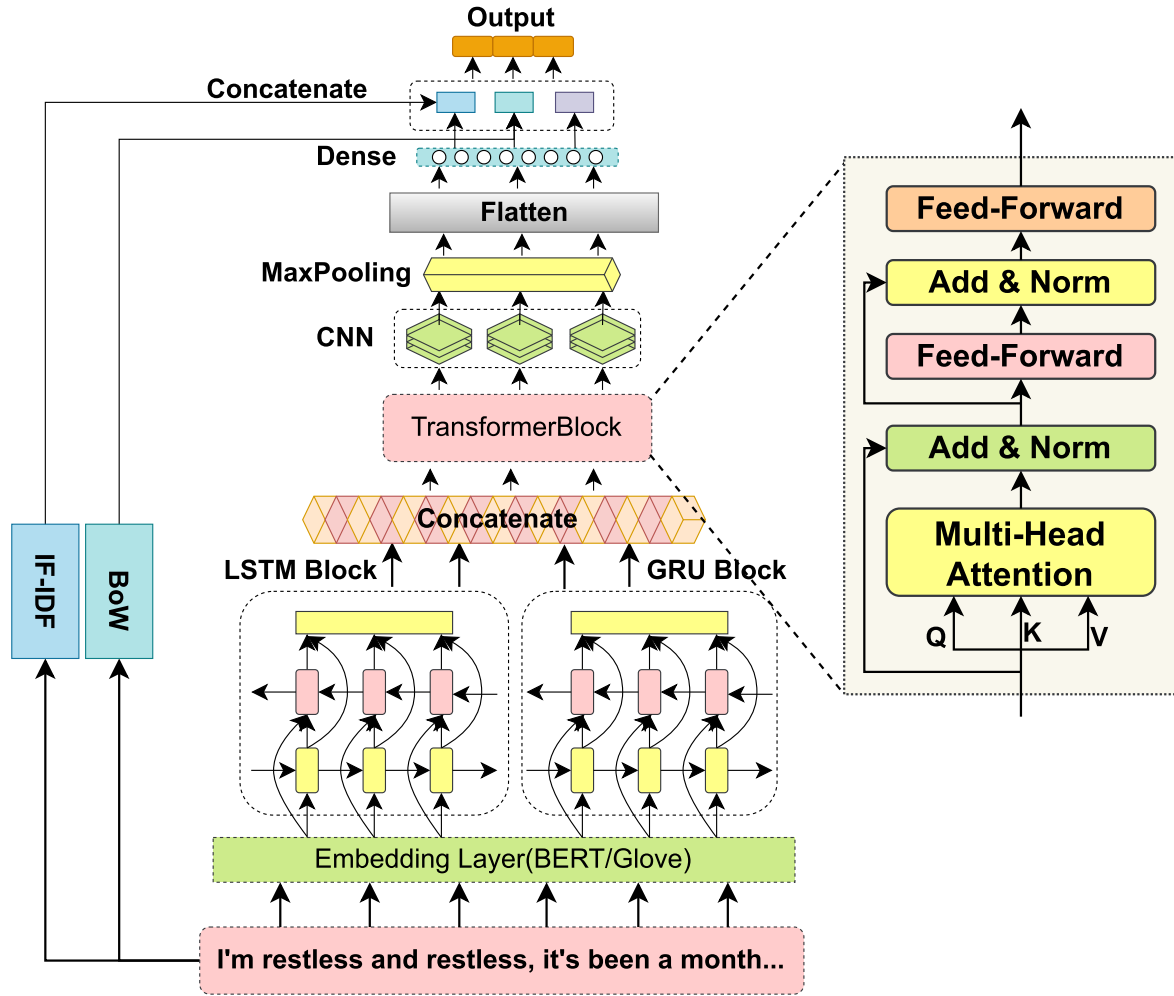


FIGURE 4. A comprehensive pipeline for mental health sentiment analysis: text embedding using BERT/Glove, Transformer blocks, MaxPooling, CNN, GRU, and LSTM, followed by flattening, dense layers, and concatenation for final output.

- **Input IDs (X_{input}):** Tokenized representations of text sequences, each with a unique identification from the vocabulary.
- **Attention Mask (A_{mask}):** These masks specify which tokens are genuine and which are padding, such that padding tokens are excluded from the attention computation.

Formally, the inputs are defined as:

$$X_{\text{input}} \in \mathbb{R}^{B \times L}, \quad A_{\text{mask}} \in \mathbb{R}^{B \times L} \quad (1)$$

where B is the batch size, and L is the maximum sequence length.

We used input IDs to encode the token sequence in each batch, and the attention mask enabled the model to focus on valid tokens while ignoring padding throughout the attention mechanism. This ensured that the padding tokens did not contribute to the model's computations, hence increasing efficiency and accuracy.

B. BERT EMBEDDING LAYER

We utilized the pre-trained BERT model to extract contextual embeddings from the input sequences. BERT used the input IDs and attention masks to generate an output embedding for each token:

$$H_{\text{BERT}} = \text{BERT}(X_{\text{input}}, A_{\text{mask}}) \quad (2)$$

Here, $H_{\text{BERT}} \in \mathbb{R}^{B \times L \times d}$, where d is the hidden size of the BERT model (768). We used these embeddings, which record both the semantic meaning and the contextual links between tokens, allowing the model to draw on pre-trained language knowledge. By feeding the input sequences along with associated attention masks, BERT was able to generate high-quality, context-aware token representations, which improved our model's grasp of the text.

BERT (Bidirectional Encoder Representations from Transformers) embeddings are dynamic representations that capture the relationships between words dependent on their context. Unlike static embeddings like Word2Vec or GloVe,

BERT embeddings adjust to the precise usage of a word in a sentence. This makes them especially useful for mental health sentiment analysis, where linguistic nuances and domain-specific terminology are important.

C. LSTM AND GRU LAYERS

To model temporal dependencies, we passed $\mathbf{H}_{\text{BERT}}$ through Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) layers.

LSTMs and GRUs are both specialized types of recurrent neural networks (RNNs) designed to handle sequential data. LSTMs use memory cells and gating mechanisms to preserve information over long sequences, making them ideal for capturing long-term dependencies. GRUs, on the other hand, simplify the LSTM architecture by combining the forget and input gates, resulting in fewer parameters and faster computation while still effectively modeling sequential dependencies. In this study, both architectures were used to complement each other, with LSTMs focusing on long-range temporal patterns and GRUs ensuring computational efficiency.

- **LSTM Layer:** This layer captured long-range dependencies by maintaining and updating the cell state over time. The LSTM equations were as follows:

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (3)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (4)$$

We employed the LSTM to capture both short-term and long-term dependencies in the sequence, taking advantage of its ability to retain essential information over long durations.

- **GRU Layer:** This layer represented interdependence with fewer parameters and a simpler gating mechanism than the LSTM. The GRU equations are as follows:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} \quad (5)$$

$$+ \mathbf{z}_t \odot \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h \mathbf{r}_t \odot \mathbf{h}_{t-1} + \mathbf{b}_h) \quad (6)$$

We chose the GRU for its efficiency, capturing dependencies with fewer computational resources while still being effective for sequence modeling.

The outputs from both layers, $\mathbf{H}_{\text{LSTM}}$ and $\mathbf{H}_{\text{GRU}}$, were then concatenated to form the final sequence representation:

$$\mathbf{H}_{\text{seq}} = \text{Concatenate}(\mathbf{H}_{\text{LSTM}}, \mathbf{H}_{\text{GRU}}) \quad (7)$$

This allowed us to combine the strengths of both architectures, enhancing our model's ability to capture both complex temporal dependencies and computational efficiency.

D. TRANSFORMER BLOCK

We refined $\mathbf{H}_{\text{seq}}$ using a Transformer block to capture complex dependencies within the sequence.

Transformers are advanced deep learning architectures that capture contextual links between sequence items through self-attention. This approach assigns varied priority to each

token, allowing the model to address both local and global dependencies. Transformers are more efficient for large datasets than recurrent models like LSTMs and GRUs since they process sequences in parallel. The transformer's multi-head attention mechanism allowed our model to pay to several parts of the sequence at the same time, computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (8)$$

Here, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ were derived from $\mathbf{H}_{\text{seq}}$, allowing the model to capture intricate relationships.

After attention, a position-wise feed-forward network further refined the sequence representation:

$$\mathbf{H}_{\text{ffn}} = \text{ReLU}(\mathbf{W}_1 \mathbf{H}_{\text{attn}} + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2 \quad (9)$$

Residual connections and layer normalization around the attention and feed-forward layers maintained gradient flow and improved training stability. This process enriched $\mathbf{H}_{\text{seq}}$ with contextually aware representations.

E. CONVOLUTIONAL AND POOLING LAYERS

The output from the Transformer block, $\mathbf{H}_{\text{transformer}}$, was further processed through a 1D convolutional layer to extract local features from the sequence. Specifically, we applied a 1D convolution operation followed by a ReLU activation function:

$$\mathbf{H}_{\text{conv}} = \text{ReLU}(\text{Conv1D}(\mathbf{H}_{\text{transformer}})) \quad (10)$$

This step allowed our model to capture spatial patterns and feature interactions within the sequence.

To reduce the dimensionality of the feature map and retain only the most significant information, we applied max-pooling:

$$\mathbf{H}_{\text{pool}} = \text{MaxPool1D}(\mathbf{H}_{\text{conv}}) \quad (11)$$

Max-pooling efficiently downsampled the feature map by choosing the highest value in each pooling window, increasing computational efficiency while maintaining key features.

This combination of convolution and pooling enhanced the sequence representations, preparing them for downstream tasks by stressing key patterns in the data.

F. DENSE LAYERS AND FEATURE INTEGRATION

Following the convolution and pooling stages, we flattened the pooled features, $\mathbf{H}_{\text{pool}}$, to prepare them for fully connected layers. These dense layers enabled the model to learn high-level abstract representations of the data. Specifically, we processed the flattened features, $\mathbf{H}_{\text{flat}}$, through a dense layer with ReLU activation and applied dropout for regularization:

$$\mathbf{H}_{\text{dense}} = \text{Dropout}(\text{ReLU}(\mathbf{W}_3 \mathbf{H}_{\text{flat}} + \mathbf{b}_3)) \quad (12)$$

The dropout mechanism prevented overfitting by randomly deactivating a fraction of neurons during training, resulting in improved generalization.

Algorithm 1 Hybrid Model With BERT Embeddings

```

1: Define Inputs:
2:  $\mathbf{X} \in \mathbb{R}^{\text{max\_seq\_length}}$ 
3:  $\mathbf{M} \in \mathbb{R}^{\text{max\_seq\_length}}$ 
4:  $\mathbf{T} \in \mathbb{R}^{1000}$ 
5:  $\mathbf{B} \in \mathbb{R}^{1000}$ 
6: Load Pretrained BERT:
7:  $\text{bert\_output} = \text{BERT}(\mathbf{X}, \mathbf{M})$ 
8:  $\text{bert\_embeddings} = \text{bert\_output.last\_hidden\_state}$ 
9: Apply LSTM and GRU:
10:  $\text{lstm\_out} = \text{LSTM}(64)(\text{bert\_embeddings})$ 
11:  $\text{gru\_out} = \text{GRU}(64)(\text{bert\_embeddings})$ 
12: Concatenate LSTM and GRU:
13:  $\text{concatenated} = \text{Concatenate}([\text{lstm\_out}, \text{gru\_out}])$ 
14: Apply Transformer Block:
15:  $\text{transformer\_out} = \text{TransformerBlock}(\text{concatenated}, \text{embed\_dim} = 128, \text{num\_heads} = 2, \text{ff\_dim} = 128)$ 
16: Apply CNN Layer:
17:  $\text{conv\_input} = \text{Conv1D}(64, \text{kernel\_size} = 3)(\text{transformer\_out})$ 
18:  $\text{maxpool} = \text{MaxPooling1D}(\text{pool\_size} = 2)(\text{conv\_input})$ 
19:  $\text{flatten} = \text{Flatten}(\text{maxpool})$ 
20: Apply Dense Layers:
21:  $\text{dense} = \text{Dense}(128, \text{activation} = 'ReLU')(\text{flatten})$ 
22:  $\text{dense} = \text{Dropout}(0.2)(\text{dense})$ 
23:  $\text{dense} = \text{Dense}(64, \text{activation} = 'ReLU')(\text{dense})$ 
24: Concatenate with TF-IDF and BoW:
25:  $\text{concat\_features} = \text{Concatenate}([\text{dense}, \mathbf{T}, \mathbf{B}])$ 
26: Apply Output Layer:
27:  $\text{output} = \text{Dense}(\text{num\_classes}, \text{activation} = 'Softmax')(\text{concat\_features})$ 
28: Compile Model:
29:  $\text{model} = \text{Model}(\text{inputs} = [\mathbf{X}, \mathbf{M}, \mathbf{T}, \mathbf{B}], \text{outputs} = \text{output})$ 
30:  $\text{model.compile}(\text{optimizer}=\text{Adam}(\text{learning\_rate}=2\text{e-}5), \text{loss}=\text{'categorical\_crossentropy'}, \text{metrics}=[\text{'accuracy'}])$ 
31: Return the compiled model

```

To improve the feature representation, we combined TF-IDF and Bag-of-Words (BoW) representations. These were added to the dense features to create the final feature vector:

$$\mathbf{H}_{\text{concat}} = \text{Concatenate}([\mathbf{H}_{\text{dense}}, \mathbf{H}_{\text{TF-IDF}}, \mathbf{H}_{\text{BoW}}]) \quad (13)$$

This concatenation enabled us to integrate the semantic richness of contextual embeddings with the statistical and lexical insights offered by traditional text representations. The generated feature vector, $\mathbf{H}_{\text{concat}}$, contained both deep contextual and surface-level textual information, making it well-suited for downstream classification.

G. OUTPUT LAYER

Finally, we passed the concatenated features, $\mathbf{H}_{\text{concat}}$, through a dense layer with a softmax activation function to complete the classification. The dense layer performed a linear transformation, and the softmax function transformed the results into a probability distribution over the target classes:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_4 \mathbf{H}_{\text{concat}} + \mathbf{b}_4) \quad (14)$$

This method assigned a probability score to each class, indicating how confident our model was in its prediction.

Using the rich feature representation in $\mathbf{H}_{\text{concat}}$, the softmax activation ensured robust sentiment classification, effectively capturing both contextual and statistical patterns in input data.

This final step allowed us to translate the learnt representations to the desired output, resulting in correct predictions for the sentiment analysis challenge.

H. ALGORITHM

Algorithm 1 describes the architecture of our BERT-Fuse model, which combines BERT embeddings with deep learning layers. The procedure starts with input sequences such as token IDs, attention masks, and TF-IDF/BoW characteristics. BERT embeddings are retrieved and analyzed using LSTM and GRU layers to detect sequential dependencies. The outputs are then passed through a Transformer block to capture long-range relationships, followed by a CNN layer to extract local features. The features are concatenated with TF-IDF and BoW data before being routed through dense layers and an output softmax layer for classification. The model is built using the Adam optimizer and categorical crossentropy loss for multi-class classification, merging deep learning and classical sentiment analysis methods.

I. BASELINES

We assembled a performance baseline for sentiment analysis by employing three pre-trained transformer models: BERT [31], DistilBERT [32], and RoBERTa [33]. This enabled us to compare and evaluate how effectively our proposed model, BERT-Fuse, performed sentiment classification tasks and captured contextual meaning. These models were chosen due to their shown performance on a variety of NLP tasks and ability to reflect complex linguistic patterns. The main baseline is a bidirectional transformer known as BERT, which provides a stable foundation for comparisons. A distilled version of BERT, known as DistilBERT, is used to assess how model size and efficiency effect performance. Finally, to study how training approaches affect sentiment analysis accuracy, RoBERTa, an upgraded version of BERT, is used.

- **BERT:** Bidirectional Encoder Representations from Transformers (BERT) uses a bidirectional transformer architecture to process text in both directions, capturing the whole context of each word. We utilized the pre-trained “bert-base-uncased” model from Hugging Face to leverage its robust contextual embeddings [31]. BERT’s pre-training included masked language modeling and next sentence prediction challenges, which allowed it to acquire sophisticated language patterns. This made BERT particularly useful for mental health sentiment analysis, which requires understanding the whole context of emotional expressions. By fine-tuning it on our domain-specific data, BERT displayed great performance in recognizing emotional tone and context in mental health texts, making it the dominant baseline model in our study.
- **DistilBERT:** DistilBERT is a compact and speedier variation of BERT. It retains much of BERT’s performance while lowering its size by 60%. We used the pre-trained “distilbert-base-uncased” model from Hugging Face to maximize computational efficiency, particularly in circumstances with low resources [32]. DistilBERT processes text bidirectionally to ensure complete contextual understanding. Despite its tiny size, it performed admirably in mental health sentiment analysis, providing a speedier and more resource-efficient alternative to BERT while still producing high quality outcomes.
- **RoBERTa:** RoBERTa (Robustly Optimized BERT Approach) is an upgraded version of BERT that improves the pre-training procedure by eliminating the next sentence prediction task and training on a larger dataset of longer sequences. We used the pre-trained “roberta-base” model from Hugging Face to leverage its optimized pre-training, which increases generalization and performance across multiple NLP tasks [33]. In mental health sentiment analysis, RoBERTa’s enhanced training enabled it to detect deeper contextual

TABLE 2. Model hyperparameters.

Hyperparameter	Option	Best
Max Sequence Length	80, 60, 120	80
Embedding Dimension	300, 768	768
LSTM Units	64, 128, 256	64
GRU Units	64, 128, 256	64
Transformer Embedding Dimension	128, 256	128
Number of Heads	2, 4, 6	2
Feed-Forward Dimension	128, 256	128
CNN filters	64, 128, 256	64
CNN kernel Size	3, 5, 7	3
Dense Units	64, 128, 256	128, 64
Dropout Rate	0.1, 0.2, 0.3, 0.5	0.2
Learning Rate	0.01, 1e-5, 2e-5	2e-5
Batch Size	8, 16, 32	8
Epochs	10, 20, 30	10

relationships in the text, resulting in more accurate sentiment predictions than BERT.

J. HYPERPARAMETER SETTINGS

The hyperparameters of the proposed BERT-Fuse model were carefully chosen to enhance performance for sentiment analysis, as shown in Table 2. To capture rich semantic features, input sequences are confined to a maximum length of 80 tokens, with BERT embeddings set to 768 dimensions. The model is intended for 3-class sentiment categorization and includes 64 units in both the LSTM and GRU layers to capture long-term and sequential dependencies. The Transformer block has an embedding dimension of 128, two attention heads, and a feed-forward dimension of 128 to capture token relationships. To learn local patterns, a CNN layer with 64 filters and a kernel size of 3 is utilized. The collected features are then processed by fully connected dense layers with 128 and 64 units. A dropout rate of 0.2 is used to regularize the model and prevent overfitting. To achieve stable convergence, the Adam optimizer [34] is utilized with a learning rate of 2e-5 and a batch size of 8 for 10 epochs. Early stopping was employed during training to prevent overfitting and ensure optimal model performance. These settings were chosen to strike a balance between computing efficiency, regularization, and model correctness when performing sentiment analysis tasks.

V. EXPERIMENT AND RESULT ANALYSIS

To promote effective processing and model training for our mental health sentiment analysis assignment, we created an experimental setting that addresses the domain’s specific requirements. We used Python 3.9 and its strong libraries for natural language processing and deep learning. Our models were mostly created and fine-tuned with TensorFlow 2.12,

which provides powerful capabilities for controlling complex neural network topologies. For hardware, we used an NVIDIA RTX 2060 12GB GPU, which allowed for efficient parallel computations and accelerated the training process, particularly when dealing with huge datasets. The system also has an Intel processor and 16GB of RAM, resulting in a well-balanced configuration for feature extraction, data preprocessing, and model validation. This arrangement ensures that the model performs optimally and efficiently throughout the development and testing phases.

A. EVALUATION METRICS

To ensure a thorough examination, we analyzed our hybrid model's performance using a variety of measures. We first translated the test data into TF-IDF and BoW features, and then used the trained model to predict the sentiment labels. The predicted and true labels were compared, and macro accuracy, precision, recall, and F1-score were calculated. These metrics were computed using the following formulas:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

- **Macro Precision:**

$$\text{Macro Precision} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \quad (16)$$

- **Macro Recall:**

$$\text{Macro Recall} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (17)$$

- **Macro F1-score:**

$$\text{Macro F1} = \frac{1}{N} \sum_{i=1}^N 2 \times \frac{\left(\frac{TP_i}{TP_i + FP_i} \right) \times \left(\frac{TP_i}{TP_i + FN_i} \right)}{\left(\frac{TP_i}{TP_i + FP_i} \right) + \left(\frac{TP_i}{TP_i + FN_i} \right)} \quad (18)$$

In addition, we generated confusion matrices to illustrate classification performance, which shows the distribution of true positives (TP), true negatives (TN), false positives (FP), and false negatives. A classification report was also produced to provide extensive information about the model's performance in several sentiment classes. We also generated and presented the ROC curve to evaluate the model's ability to distinguish between classes at different thresholds, examining the trade-off between sensitivity and specificity.

This evaluation provides a fully-rounded assessment of the hybrid model's performance, focusing on overall accuracy as well as class-specific precision, recall, and F1-score for a complete knowledge of its behavior.

B. EXPERIMENT WITH DIFFERENT EMBEDDING

To craft an effective sentiment analysis model, we integrated three different word embedding approaches (BERT, GloVe, and FastText) into our hybrid architecture, BERT-Fuse. Using these embedding techniques, we were able to extract a

TABLE 3. Comparison of different embedding techniques.

Embedding Technique	Accuracy	Precision	Recall	F1
With BERT	97.05%	97.38%	97.27%	97.32%
With GloVe	86.46%	86.90%	86.42%	86.50%
With FastText	84.36%	84.80%	84.40%	84.59%
Without Embedding	95.63%	95.63%	95.63%	95.63%

significant quantity of syntactic and semantic information from the text. We used GloVe to capture global word relationships, BERT embeddings to take advantage of its contextual awareness, and FastText to handle morphological nuances and terms not found in the dictionary. BERT-Fuse was taught and evaluated using each of these embedding approaches separately. Finally, we discovered that the model using BERT embeddings consistently outperformed the GloVe and FastText versions, achieving higher F1-scores, accuracy, precision, and recall throughout the sentiment classification tests. This demonstrates how well BERT's contextualized representations can detect text's nuanced sentiment cues.

Table 3 compares word embedding approaches (BERT, GloVe, and FastText) incorporated within the BERT-Fuse architecture. The results show that the model using BERT embeddings regularly outperforms the alternatives, with the highest accuracy (97.05%) and F1-score (97.32%). This demonstrates BERT's exceptional ability to extract contextual information and complex sentiment cues. The GloVe and FastText embeddings had much poorer performance, with F1-scores of 86.50% and 84.59%, respectively, suggesting their limits in handling complex sentiment expressions. We examined the model without word embeddings and obtained a competitive F1-score of 95.63%, indicating that other architectural components of BERT-Fuse contribute effectively to sentiment categorization. However, including BERT embeddings improved performance even more, highlighting the value of contextualized representations in sentiment analysis.

C. BASELINE COMPARISON

Table 4 presents a performance comparison of three baseline models (BERT, DistilBERT, RoBERTa) and the proposed BERT-Fuse model, evaluated with and without augmentation. Augmentation consistently improves the performance of all models across all metrics, highlighting its significance in enhancing generalization. For instance, BERT achieves a test accuracy of 93.01% and an F1-score of 93.70% with augmentation, compared to a test accuracy of 91.34% and an F1-score of 92.45% without augmentation. Similarly, DistilBERT achieves notable improvements with augmentation, achieving a test accuracy of 93.08% and an F1-score of 93.85%, compared to 91.34% and 92.24%, respectively, without augmentation.

TABLE 4. Baseline performances comparison with and without augmentation.

Model	Augmentation	Training Accuracy	Val Accuracy	Test Accuracy	Precision	Recall	F1-score
BERT	Yes	89.15%	93.15%	93.01%	93.97%	93.46%	93.70%
	No	91.53%	91.14%	91.34%	92.58%	92.42%	92.45%
DistilBERT	Yes	88.94%	92.91%	93.08%	94.07%	93.71%	93.85%
	No	85.09%	90.83%	91.34%	92.20%	92.31%	92.24%
RoBERTa	Yes	67.19%	78.77%	78.58%	80.54%	78.57%	79.14%
	No	57.28%	72.78%	72.93%	72.98%	75.61%	73.37%
BERT-Fuse	Yes	99.01%	96.92%	97.05%	97.38%	97.27%	97.32%
	No	96.57%	91.97%	91.76%	92.97%	92.46%	92.70%

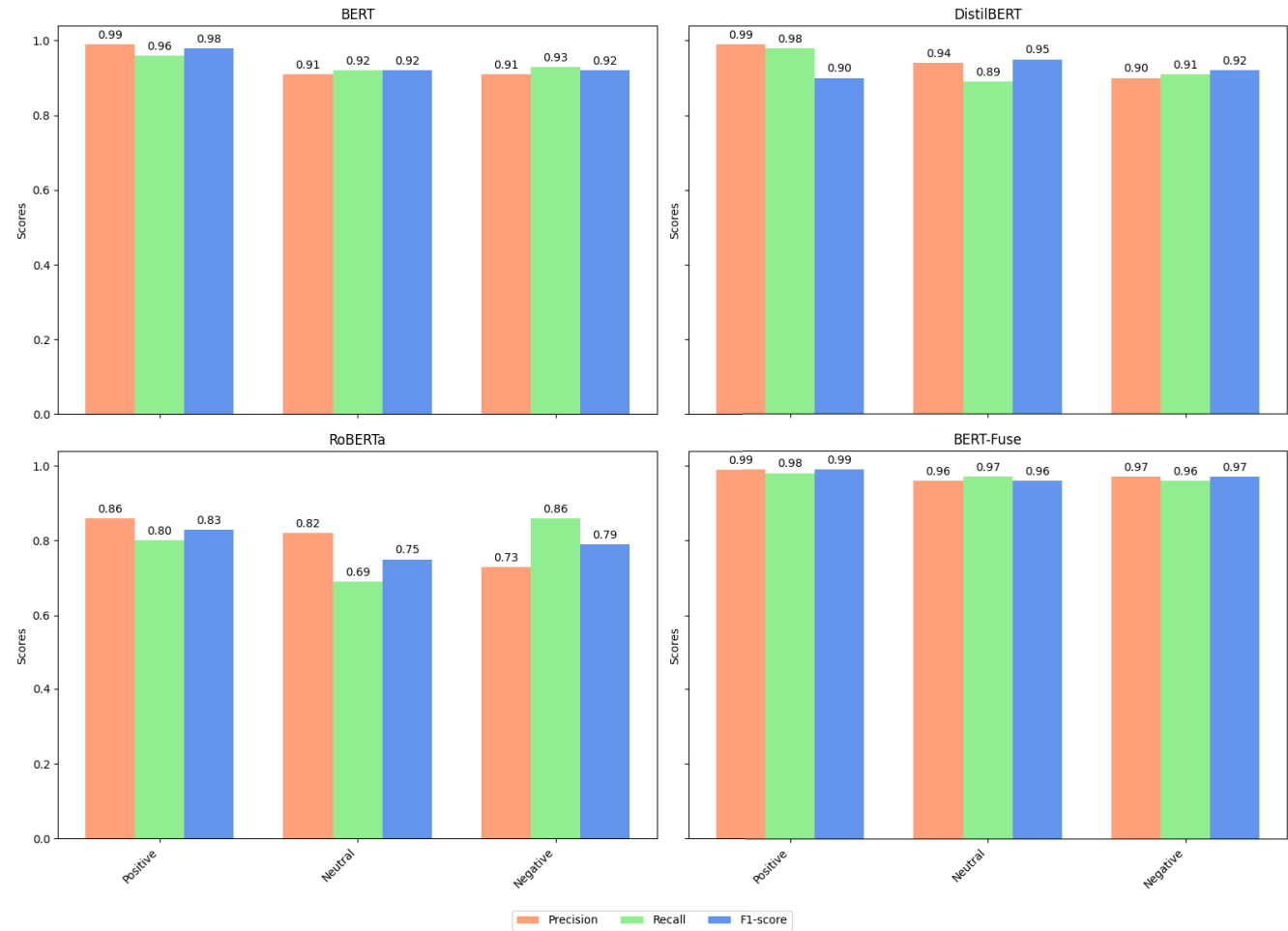


FIGURE 5. Class-wise baseline comparison of different models.

RoBERTa demonstrates the weakest performance among all models, achieving a test accuracy of 78.58% and an F1-score of 79.14% with augmentation, and 72.93% test accuracy and 73.37% F1-score without augmentation. This indicates that while RoBERTa benefits from augmentation, its overall performance is significantly lower than the other models. On the other hand, the proposed BERT-Fuse model

outperforms all baseline models, achieving the highest test accuracy of 97.05% and an F1-score of 97.32% with augmentation. Without augmentation, BERT-Fuse still demonstrates strong performance with a test accuracy of 91.76% and an F1-score of 92.70%, surpassing the other models.

Across all models, augmentation improves precision, recall, and F1-score values. BERT-Fuse frequently

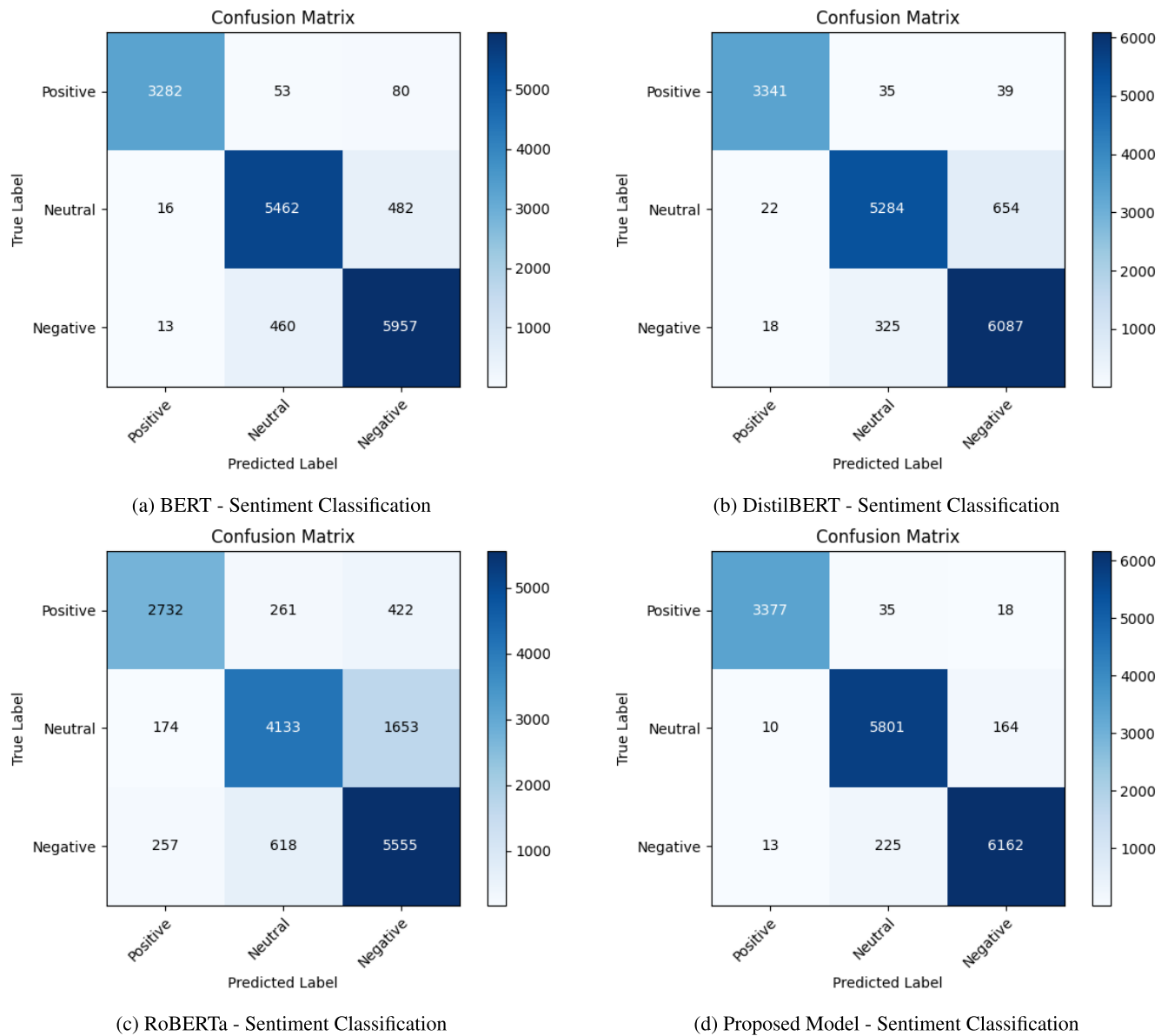


FIGURE 6. Confusion matrices for sentiment classification using different models.

outperforms baseline models, demonstrating that its architecture is more effective at capturing data subtleties. Even without augmentation, BERT-Fuse produces outcomes equivalent to or better than augmented baseline models, demonstrating its robustness. Not classified. However, including BERT embeddings improved performance even more, highlighting the value of contextualized representations in sentiment analysis.

Figure 5 presents the class-wise precision, recall, and F1-score, illustrating how each model performs on individual classes. The proposed BERT-Fuse model maintains high scores across all classes, demonstrating its ability to handle class imbalances effectively and outperform baseline models in fine-grained classification. Overall, augmentation plays a crucial role in improving model performance, but the

BERT-Fuse model remains the most effective across all metrics, establishing itself as the best-performing model in this evaluation.

1) CONFUSION METRICS

The confusion matrices in Figure 6 illustrate the performance differences among the models for sentiment classification. Figure 6a shows that the BERT model achieves a balanced distribution of predictions but leaves room for improvement in reducing misclassifications. The majority of misclassifications occur in the neutral class, where 482 neutral instances were misclassified as negative. This indicates that while BERT is effective in sentiment classification, it struggles slightly in differentiating neutral sentiments.

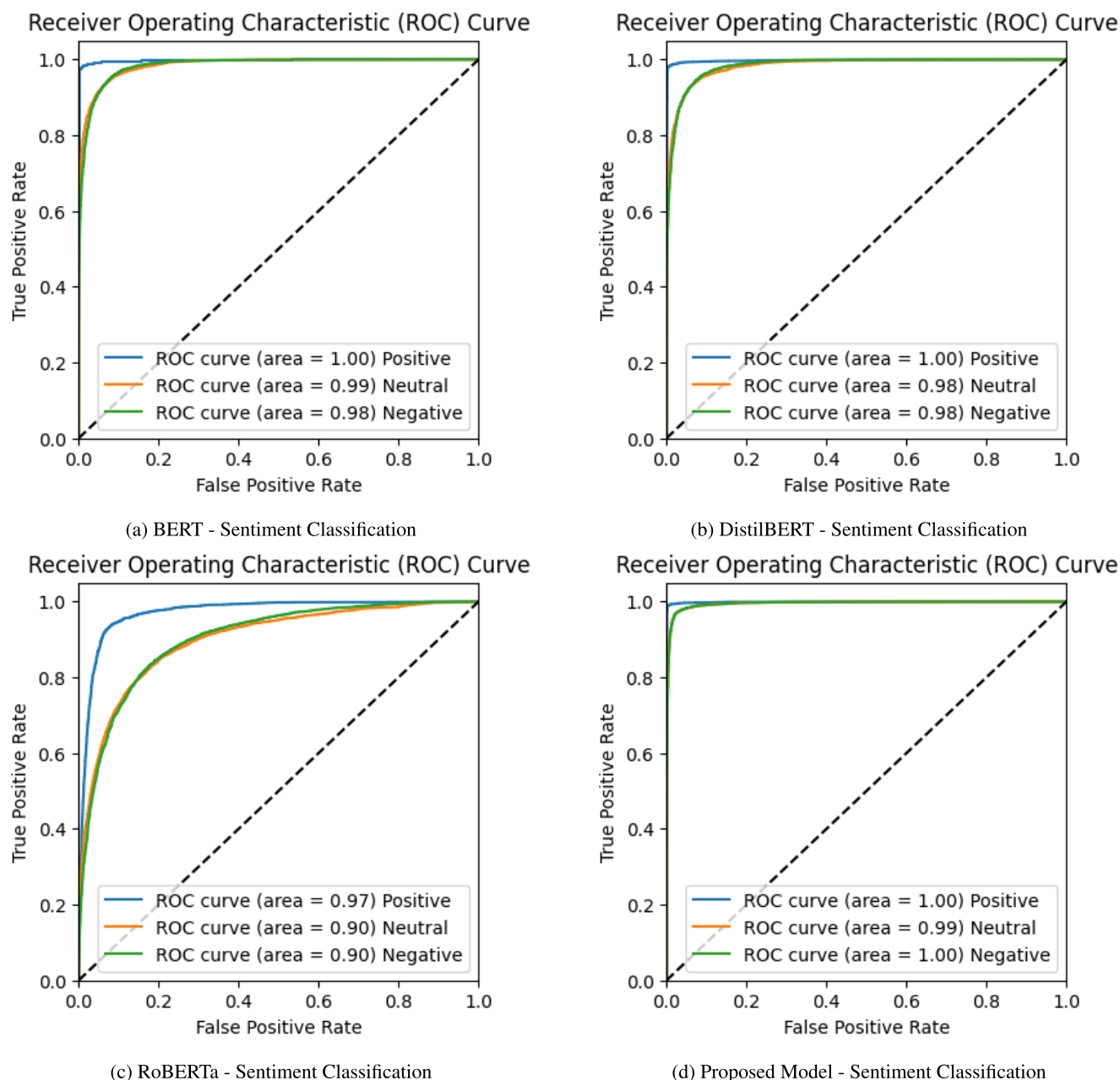


FIGURE 7. ROC Curves for sentiment classification using different models.

Figure 6b demonstrates that DistilBERT improves slightly over BERT with a marginal reduction in misclassification rates. Specifically, DistilBERT classifies 5284 neutral instances correctly, compared to 5462 for BERT, but it reduces the number of neutral instances misclassified as negative to 654 (down from 482 in BERT). Moreover, the number of positive instances misclassified as neutral is lower (35 compared to 53 in BERT), indicating that DistilBERT refines sentiment classification performance, particularly in distinguishing between positive and neutral sentiments.

Figure 6c shows that RoBERTa, despite its strengths, misclassifies more instances compared to DistilBERT. It performs relatively well in detecting negative sentiments (5555 correctly classified), but it struggles in neutral classifications, with 1653 neutral instances being misclassified as negative. Additionally, the number of misclassified positive instances is higher than in BERT and DistilBERT, with 261 positive instances classified as neutral and 422 as negative.

Finally, Figure 6d shows that the proposed BERT-Fuse model outperforms all baseline models, resulting in more correct classifications and fewer misclassifications. The

TABLE 5. Ablation study results.

Model Variation	Test Accuracy	Precision	Recall	F1-Score
BERT-Fuse	97.05%	97.38%	97.27%	97.32%
Without GRU	94.60%	95.28%	94.95%	95.11%
Without Transformer	94.31%	95.04%	94.85%	94.91%
Without CNN	94.76%	95.50%	95.16%	95.29%
Without LSTM	94.70%	95.26%	95.20%	95.23%
Without TF-IDF	96.14%	96.52%	96.43%	96.47%
Without BoW	73.35%	74.70%	74.16%	74.31%
Without Both (TF-IDF & BoW)	95.98%	96.47%	96.25%	96.36%

suggested methodology properly detects 5801 neutral cases, greatly lowering the number of incorrectly categorized neutral attitudes. Furthermore, it minimizes false positives and negatives across all emotion classes, demonstrating its efficacy in increasing sentiment classification accuracy. The number of positive examples misclassified as neutral is decreased to 35, while the number of neutral instances misclassified as negative drops to 164, demonstrating the suggested model's greater generalization capabilities.

These results emphasize the effectiveness of the proposed model in enhancing sentiment classification accuracy by improving correct classifications and minimizing misclassification errors across different sentiment categories.

2) ROC CURVE ANALYSIS

Figure 7 shows the ROC curves for BERT, DistilBERT, RoBERTa, and the proposed BERT-Fuse model, which are used for sentiment classification. The ROC curve examines the trade-off between true positive and false positive rates, providing information about each model's overall classification capabilities.

In Figure 7a, the BERT model performs well with high Area Under the Curve (AUC) values for all three sentiment classes: positive (1.00), neutral (0.99), and negative (0.98). These results show that BERT achieves near-perfect categorization for positive sentiments while still performing well for neutral and negative classes. However, the somewhat lower AUC for neutral categorization indicates some misclassification concerns, as shown in the confusion matrix. Figure 7b demonstrates that DistilBERT works well, with AUC values of 1.00 for positive sentiments and 0.98 for neutral and negative feelings. While DistilBERT falls slightly behind BERT in neutral sentiment classification, the total performance remains very competitive. The minor drop in the AUC for neutral and negative classes implies that, while DistilBERT is more efficient, it may forfeit accuracy when compared to the original BERT model. Figure 7c shows that RoBERTa has lower AUC values, especially for neutral and negative feelings (0.90). The lower AUC values show that RoBERTa performs worse than the other models, notably in distinguishing between neutral and negative feelings. This is consistent with the confusion matrix data, which showed that RoBERTa had the most misclassified cases. Despite a 0.97 AUC for positive sentiment, it performs substantially worse than the other classes. Finally, Figure 7d shows that

the proposed BERT-Fuse model has the highest AUC values across all classes, with perfect scores of 1.00 for positive and negative sentiments and 0.99 for neutral sentiment. This result demonstrates the proposed model's robustness, as it beats baseline approaches in sentiment classification tasks. The BERT-Fuse model effectively decreases false positives and negatives, resulting in more accurate classification across sentiment categories.

These findings highlight the usefulness of the proposed BERT-Fuse model, which consistently achieves high classification accuracy while minimizing misclassification errors across all sentiment categories.

D. ABLATION STUDY RESULTS

We carried out an ablation investigation on the hybrid BERT-Fuse model to determine the contribution of each component to overall performance. The BERT-Fuse model combines several critical components, including as BERT embeddings, GRU, Transformer, CNN, and TF-IDF/BoW characteristics. To assess the value of these components, we removed one at a time, resulting in different models. Each variation was trained and evaluated, and its performance indicators, such as testing accuracy, macro-averaged precision, recall, and F1-score, were compared to the overall BERT-Fuse model. The ablation study results are summarized in Table 5. The entire BERT-Fuse model had the highest performance across all metrics, with a testing accuracy of 97.05%, macro-averaged precision of 97.38%, macro-averaged recall of 97.27%, and F1-score of 97.32%. Removing the GRU reduced testing accuracy to 94.60% and F1-score to 95.11%, highlighting the GRU's crucial role in detecting sequential dependencies. Similarly, eliminating the Transformer caused a comparable reduction, with a testing accuracy of 94.31% and an F1-score of 94.91%. This highlights the relevance of the Transformer's capacity to represent long-range interactions. The CNN component also significantly boosted the model's performance. When it was excluded, the testing accuracy declined to 94.76% and the F1-score to 95.29%, indicating its effectiveness in extracting local characteristics. Interestingly, eliminating the LSTM had a small impact, with the model still obtaining a testing accuracy of 94.70% and an F1-score of 95.23%. This suggests some redundancy in this component given the inclusion of the GRU. The study also underlined the importance of feature representations. deleting TF-IDF

TABLE 6. Comparison of various approaches for mental health detection and sentiment analysis.

Citation	Dataset	Problem Statement	Proposed Methodology	Accuracy
Bao et al. [17]	Text data	Identifying complex emotional expressions in textual data to improve sentiment analysis models for mental health applications.	TCSO-DGNN (Hybrid graph neural network)	>93% (Precision, Recall, F1-score)
Zhang et al. [20]	Text data	Detecting individuals at risk of depression by analyzing linguistic patterns and emotional cues in textual data.	Hierarchical Transformer Network with retrieval-based sampling	95.46%
Ghosh et al. [28]	Bangla social media text	Analyzing Bangla social media conversations to detect depressive sentiments and assess mental health status.	BiLSTM + CNN	94.3%
Alanazi et al. [13]	Financial news texts	Extracting sentiment for public mental health monitoring by leveraging NLP and deep learning techniques to analyze emotional patterns and psychological trends in textual data.	CNN	93.9%
S. Almutairi [27]	Text-based interaction dataset (young users)	Early depression detection using hybrid models with attention mechanisms.	CNN + TS-LSTM with feature attention	97.23%
Kour et al. [29]	Twitter data	Classifying depression-related tweets by identifying linguistic features indicative of depressive tendencies.	CNN + BiLSTM	94.28%
Kaushik et al. [9]	Social network data	Classifying mental health statuses from extensive, preprocessed datasets by leveraging deep learning and NLP techniques to capture emotional nuances and contextual meanings.	Ensemble methods (XGBoost)	82%
Our Work	Mental Health	The aim of BERT-Fuse is to advance the field of mental health sentiment analysis by integrating traditional feature extraction methods with state-of-the-art deep learning architectures. The model seeks to facilitate accurate and scalable detection of mental health conditions through the nuanced analysis of domain-specific textual data.	BERT-Fuse	97.05%

resulted in a testing accuracy of 96.14% and an F1-score of 96.47%. However, deleting BoW resulted in a significant performance reduction, with the testing accuracy plunging to 73.35% and the F1-score to 74.31%. This large drop suggests that BoW is critical for offering complementary features. When TF-IDF and BoW were removed, the model retained reasonable performance, with a testing accuracy of 95.98% and an F1-score of 96.36%. This demonstrates that while these elements boost the model’s performance, the underlying architecture remains robust. In summary, the ablation investigation highlighted the significance of each component in the BERT-Fuse model. Components such as GRU, Transformer, CNN, and BoW were highly important in achieving high performance, while removing TF-IDF or LSTM had a smaller influence. This study

provides useful insights into the contributions of various components and identifies areas for future improvement or simplification.

E. COMPARISON AMONG AND EXISTING WORK

Table 6 presents a comparative analysis of various existing approaches for mental health detection and sentiment analysis, highlighting their datasets, problem statements, methodologies, and reported accuracy. Several studies have leveraged traditional machine learning and deep learning techniques to address mental health-related challenges using social media and curated datasets. Bao et al. [17] used a hybrid graph neural network (TCSO-DGNN) for sentiment analysis in text data, achieving >93% accuracy. Zhang et al. [20] employed a Hierarchical Transformer

Network for depression risk detection, reaching 95.46% accuracy. Ghosh et al. [28] analyzed Bangla social media conversations with a BiLSTM-CNN model (94.3%), while Alanazi et al. [13] applied CNN to financial news texts (93.9%).

For early depression detection, Almutairi [27] combined CNN with TS-LSTM and feature attention, achieving 97.23%. Kour et al. [29] classified depression-related tweets using CNN-BiLSTM (94.28%). Kaushik et al. [9] used ensemble methods (XGBoost) for social network data classification, obtaining 82

Compared to these approaches, our proposed BERT-Fuse model integrates traditional feature extraction techniques, such as TF-IDF and BoW, with deep learning architectures, including BERT embeddings, CNN, GRU, Transformer blocks, and LSTM layers. This hybrid approach enhances the model's ability to capture nuanced sentiment patterns in mental health-related texts, leading to an improved accuracy of 97.05%. These results underscore the effectiveness of BERT-Fuse in advancing mental health sentiment analysis by combining contextualized embeddings with diverse learning paradigms for scalable and precise detection of mental health conditions.

VI. DISCUSSION

BERT-Fuse has enhanced mental health sentiment analysis by effectively capturing nuanced emotional cues in mental health texts. By combining BERT's contextual embeddings with CNN for local feature extraction and LSTM/GRU for sequential dependencies, the model offers a robust approach to understanding emotional and behavioral signals, improving performance for early detection and intervention in mental health applications. A key finding of this study is the significant impact of BERT embeddings combined with domain-specific features. The pre-trained BERT embeddings allowed the model to capture contextual information, outperforming traditional models in terms of accuracy, precision, recall, and F1-score. BERT proved to be the most effective in identifying subtle sentiment cues in the mental health domain, highlighting the value of context-aware embeddings for tasks involving complex emotional expression.

The hybrid structure of BERT-Fuse, integrating CNN, LSTM, and GRU, played a critical role in performance. CNN layers captured local patterns such as n-grams that indicate sentiment shifts, while LSTM/GRU layers accounted for long-range dependencies, crucial for tracking emotional progression in mental health texts. This combination of local and global feature extraction is vital for understanding complex linguistic structures, tone variations, and subtle emotional cues. The multi-task learning framework enabled BERT-Fuse to predict multiple sentiment-related tasks simultaneously, improving generalization across different mental health conditions while reducing overfitting. This shared learning approach resulted in superior performance on all evaluation metrics, demonstrating the effectiveness of multi-task learning in mental health sentiment analysis.

Additionally, the data augmentation technique, particularly synonym replacement, helped improve the model's robustness by expanding the dataset and introducing linguistic diversity. This, in combination with both traditional and deep learning features, allowed BERT-Fuse to generalize well across a variety of mental health texts. The ablation study further emphasized the complementary nature of BERT-Fuse's components. Removal of any feature, whether GRU, CNN, Transformer blocks, or traditional TF-IDF/BoW, resulted in a drop in performance, reinforcing the importance of each element in the model. Overall, BERT-Fuse offers a promising approach to mental health sentiment analysis by integrating advanced NLP techniques with traditional feature extraction methods. The model's ability to capture nuanced emotional and behavioral cues makes it a valuable tool for accurate, scalable, and effective mental health detection, enabling more timely and appropriate interventions.

VII. LIMITATION AND FUTURE WORK

Despite the promising results obtained by the BERT-Fuse model, certain limitations must be acknowledged. First, the quality and diversity of the dataset have a substantial influence on the model's performance. Although the 50,000-sample dataset is huge, the model may struggle to generalize if the data does not reflect all demographic categories or mental health conditions. While using many layers such as GRU, Transformer, and CNN is useful for capturing complex emotion factors, it can also lead to overfitting, particularly when dealing with tiny or imbalanced datasets. Furthermore, while BERT embeddings yielded improved results, the model's findings remained difficult to grasp due to the complicated architecture, which may mask the reasons behind individual predictions.

To increase BERT-Fuse's generalizability and robustness, future research should concentrate on augmenting and diversifying training data, ensuring representation across diverse populations and mental health concerns. Other embedding strategies, multi-task learning approaches, and the incorporation of external knowledge sources such as domain-specific ontologies could all be investigated further for model optimization purposes. Research on model explainability could also help provide additional insight into how the model makes decisions. Finally, adapting BERT-Fuse to real-world scenarios, such as social media or healthcare platforms, may help to improve the model's adaptability and performance, making it a more useful tool for mental health detection and treatment.

VIII. CONCLUSION

Mental health detection is critical for timely intervention, yet traditional methods face challenges due to the complexity of mental health-related texts. Sentiment analysis, powered by natural language processing (NLP), is a potential way for detecting emotional cues in these writings. These complexities have been successfully addressed by hybrid models, which combine sophisticated deep learning techniques with

traditional feature extraction. Our BERT-Fuse model combines multiple approaches to detect mental health attitudes with 97.05% accuracy. Even though some challenges remain, such as the need for diverse training data and improving model interpretability, by refining and applying models like BERT-Fuse, we can open up new avenues for early mental health detection, empowering proactive care, and improving lives on a global scale.

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CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest related to this research work.

DATA AVAILABILITY STATEMENT

Dataset Available at <https://shorturl.at/LzeMH>

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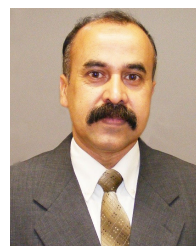
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